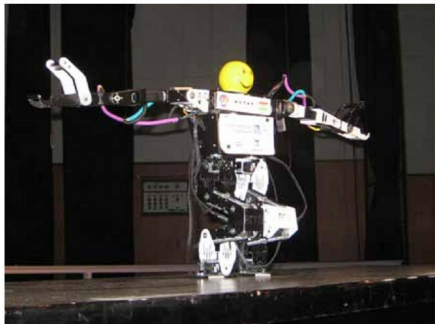




Dance demo by AcYut I

try manager, Energid Robotics and Machine Vision, gives us another example of the new roles robots are taking up: "The world's most advanced 43 degrees-of-freedom humanoid 'Robonaut' by NASA is expected to be used for replacing humans in space to carry out the maintenance tasks. It is equipped with a deeper and wider range of sensors and most advanced high-resolution cameras and infrared systems. The two Robonaut hand/wrist modules alone contain 84 sensors for feedback and control, 60 of which are analogue and require signal conditioning and digitisation. Each degree of freedom has a motor position sensor, a joint force sensor and a joint absolute position sensor. The two arm modules contain 90 sensors, 80 of which are analogue. Each actuator contains a motor incremental position sensor, redundant joint torque sensors, redundant joint absolute position sensors, and four temperature sensors distributed throughout the joint. All of these are essential to perform human-like actions." Energid, USA, has been awarded a contract by NASA to control Robonaut in distance control mode.

Very advanced, and quite what is expected of an organisation like NASA. But not as touching as the blog posted

by a team of BITS-Pilani students in June 2010: "We saw dreams, lived them and conquered them... won India's first-ever gold medal in ROBOGAMES 2010, not to forget two silvers and a bronze too!" There were plenty of challenges—lack of funds, difficulty in acquiring components, etc—but nothing stopped Samay Kohli and Arpit Mohan from making India's first indigenous humanoid robot, AcYut, for the world-renowned ROBOGAMES contest in 2008. Their team grew in size, and AcYut kept improving.

AcYut now runs on a 600MHz, single-board, Linux-based Gumstix computer that uses Texas Instruments' OMAP3503 ARM microprocessor. It uses a kind of custom-built operating system, communicates with the PlayStation 3 controller using Bluetooth, and is programmed using a completely custom-built multi-platform graphical user interface (GUI) that runs a full inverse kinematics engine together with real-time load calculations. AcYut uses motors with 106kg-cm torque and moves using spline-based trajectory generation. At ROBOGAMES last year, it danced, sumo wrestled, took part in Kung-Fu tournaments and did much more than winning hearts!

Experiments and models have shown that robots can now be made to do many, many things that we do. Having undoubtedly proved their worth in controlled environments like factories, they are now—ever so slowly—trying to mingle with the big, mean, real world outside.

"Robots have been growing rapidly and finding a place in surveillance, defence organisations, navigation systems and exploration vehicles. But the trend is now in favour of systems with greater autonomy and perfection, which can help humans in their daily activities as well as in environments such as cafés, offices, industries and hospitals. Robots such as Nexi, ASIMO and TOPIO can serve at restaurants, play ping-pong with humans, respond and communicate well with us. Advancements in new sensory technologies such as high-rate cameras, laser sensors, better computers and processors have made the path smoother for such developments," says an excited N.H. Shekar—one amongst a group of robotics enthusiasts being groomed by Dr K.G. Srinivasa, mentor-RoboMSR, M.S. Ramaiah Institute of Technology, Bengaluru.

Robot waiters and maids are just the beginning. Scientists are making robot 'babies' that begin as a clean slate and learn everything in the course of time, just like humans do. There are robot nurses that can—to a limited extent—understand their patients and even speak to them. Companies are trying to make completely autonomous, robotic trucks, cars and aeroplanes. Some hospitals and restaurants have employed robots, and so have a few homes in countries like Japan.

However, it might be decades before there is a robotic maid in every house, or a robot waiter in every restaurant, or an airport that is completely mopped by robots. Japan, considered to be one of the most technologically advanced nations, hopes to commercialise service robots on a large scale only by 2025. So it might be even later when we see such commercialisation in India. And there exists the possibility